



GOVERNMENT HOSPITAL PLUS

A Digital Medical Records Management System for Government Hospitals of Sri Lanka

Software Requirements Specification (SRS)

PRESENTED BY

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TABLE OF CONTENTS

1. Introduction	4
1.1 Project Title	4
1.2 Purpose of the System	4
1.3 Scope of the System	4
1.4 Definitions and Abbreviations	4
2. Overall Description	5
2.1 Product Perspective	5
2.2 Product Functions	5
2.3 User Classes and Characteristics	5
2.4 Operating Environment	6
2.5 Assumptions and Dependencies	6
3. Functional Requirements	6
3.1 Authentication Module	6
Description	6
Inputs	6
Outputs	6
Features	6
3.2 Patient Registration Module	6
Description	6
Inputs	6
Outputs	7
Features	7
3.3 Patient Dashboard Module	7
Description	7
Features	7
3.4 Medical Staff Module	7
Description	7
Features	7
Inputs	7
Outputs	7
3.5 Administrator Module	8
Description	8
Features	8
3.6 Medical Record PDF Export Module	8
Description	8
Features	8
3.7 Audit Log Module	8
Description	8

Features	8
3.8 Contact / Support Module	8
Description	8
Inputs	8
Outputs	8
4. Non-Functional Requirements	9
4.1 Performance Requirements	9
4.2 Security Requirements	9
4.3 Usability Requirements	9
4.5 Maintainability Requirements	9
4.6 Portability Requirements	9
5. System Requirements	10
Hardware Requirements	10
Software Requirements	10
6. Database Design	10
6.1 Database Overview	10
6.2 Core Tables	10
6.3 Relationships	11
7. User Interface Requirements	11
7.1 Design Theme	11
7.2 Screens Included	11
8. Advantages of the System	11
9. Future Enhancements	12
10. Conclusion	12

1. Introduction

1.1 Project Title

Government Hospital Plus (GH+) — A Digital Medical Records Management System for Government Hospitals of Sri Lanka

1.2 Purpose of the System

The purpose of GH+ is to digitize and centralize patient medical records across Sri Lanka's government hospital network. The system replaces paper-based record keeping with a secure, web-based platform accessible to patients, medical staff, and administrators. It aims to eliminate lost records, reduce duplicate entries, prevent medication errors, and empower patients with transparent access to their own health data.

1.3 Scope of the System

GH+ is a locally hosted, role-based web application built using PHP and MySQL (XAMPP environment). The system provides:

- Secure NIC-based authentication for three user roles: Patient, Doctor/Medical Staff, and Administrator
- Patient registration and administrative approval workflow
- Full medical record management including diagnoses, prescriptions, notes, and multi-format file attachments
- Patient-facing dashboard with read-only access to personal medical history
- PDF export of individual medical records
- Mobile-responsive interface accessible on smartphones and tablets

The system is designed specifically for deployment in Sri Lanka's government hospital context, operating on a Local Area Network (LAN) without dependency on cloud infrastructure.

1.4 Definitions and Abbreviations

Term / Abbreviation	Definition
GH+	Government Hospital Plus — the system being specified
SRS	Software Requirements Specification
NIC	National Identity Card (used as unique user identifier)
EHR	Electronic Health Record
RBAC	Role-Based Access Control
SLMC	Sri Lanka Medical Council
PHP	Hypertext Preprocessor — server-side scripting language
MySQL	Relational database management system
XAMPP	Cross-platform local server environment (Apache + MySQL + PHP)
LAN	Local Area Network

Term / Abbreviation	Definition
BCRYPT	Password hashing algorithm used for secure credential storage
MIME	Multipurpose Internet Mail Extensions — used for file type validation
Admin	System Administrator — manages users, approvals, and audit logs

2. Overall Description

2.1 Product Perspective

GH+ is a standalone web-based system that operates on a local Apache/MySQL server (XAMPP). It requires no external cloud services and is designed for intranet deployment within a government hospital network. All data is stored in a centralized MySQL database (gh_plus), and file uploads are managed on the local server filesystem. The system replaces and significantly improves upon the existing paper-based records management used by Sri Lanka's government hospitals.

2.2 Product Functions

The main functions of GH+ include:

- User authentication and role-based session management
- Patient self-registration with administrative approval gate
- Medical staff patient search (by NIC, name, city, or blood type)
- Medical record creation with multi-format file attachments
- Patient access to personal health history in read-only mode
- PDF generation and download of individual medical records
- Administrator control panel: user management, approvals, audit log, reporting
- Automatic age calculation from date of birth
- Complete audit log of all user actions

2.3 User Classes and Characteristics

User Type	Description	Technical Proficiency
Patient	Registers online; views own medical history, prescriptions, and documents; downloads PDF records	Low to Moderate
Doctor / Medical Staff	Searches patients; uploads new medical records, diagnoses, prescriptions, and file attachments	Moderate
Administrator	Full system control: approves/rejects/edits/deletes users, manages all records, views audit logs and reports	Moderate to High

2.4 Operating Environment

GH+ operates in the following environment:

- Server: XAMPP (Apache HTTP Server + MySQL+ + PHP) on Windows
- Network: Local Area Network (LAN) for multi-user access within a hospital
- Client: Any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge)
- Device Support: Desktop computers, laptops, tablets, and smartphones (mobile-responsive design)

2.5 Assumptions and Dependencies

- The system assumes a functioning LAN environment within the hospital
- XAMPP or equivalent local server stack is installed and maintained by hospital IT staff
- All users have a valid Sri Lanka National Identity Card (NIC) number for registration
- Users have access to a web browser on a connected device

3. Functional Requirements

3.1 Authentication Module

Description

Provides secure NIC-based login for all user roles. Protects all pages with session-based authentication and redirects unauthenticated access.

Inputs

- NIC number (unique identifier)
- Password

Outputs

- Authenticated session with role-specific dashboard access
- Error message on invalid credentials

Features

- BCrypt password hashing via PHP password_hash()
- Session-based access control on all protected pages
- Role detection: Patient, Doctor/Medical Staff, or Administrator
- Logout functionality with session destruction
- Audit log entry on every login and logout

3.2 Patient Registration Module

Description

Allows new patients to self-register online. Registration is held in a pending state until an Administrator approves the account.

Inputs

- Full name, NIC number, date of birth, gender
- Blood type, known diseases, additional medical notes
- Contact number, address, city
- Password (hashed on submission)

Outputs

- Pending account created in the users and patient_details tables
- Confirmation message advising the patient to await approval

Features

- NIC uniqueness validation (prevents duplicate accounts)
- Automatic age calculation from date of birth
- Patient cannot log in until the Administrator approves the account

3.3 Patient Dashboard Module**Description**

Provides the authenticated patient with a personal health dashboard showing their profile, complete medical history, prescriptions, and uploaded documents.

Features

- Display of personal profile information (name, NIC, age, blood type, diseases)
- Chronological list of all medical records (date, doctor, diagnosis, prescription)
- View attached documents per record (images, PDFs, audio, video)
- Download any individual medical record as a formatted PDF
- Contact support via the Contact Us form
- Read-only access — patients cannot modify any records

3.4 Medical Staff Module**Description**

Allows authenticated doctors and medical staff to search for patients and manage their medical records.

Features

- Patient search by NIC, full name, city, or blood type
- View full medical history and documents for any located patient
- Create new medical records including: date, diagnosis, prescription text, and additional notes
- Attach files to records: images (JPG, PNG), PDFs, audio (MP3), and video (MP4)
- File upload with extension and MIME type validation
- All uploads and record creations are logged in the audit_log table

Inputs

- Patient search criteria (NIC / name / city / blood type)
- Medical record fields: diagnosis, prescription, notes, date
- File attachments

Outputs

- Patient search results list
- Patient medical history view
- Confirmed new medical record stored in medical_records and medical_documents tables

3.5 Administrator Module

Description

Provides the Administrator with full system control including user lifecycle management, record oversight, audit log review, and reporting.

Features

- View all pending patient registrations and approve or reject them
- Edit or delete any user account (patient or staff)
- Generate summary reports on system activity and user counts
- View all medical records across the system

3.6 Medical Record PDF Export Module

Description

Allows patients to download their individual medical records as a formatted PDF document.

Features

- Auto-triggers browser print dialog pre-formatted for PDF output
- Formatted layout includes: patient name, NIC, record date, doctor, diagnosis, prescription, and notes
- Print-specific CSS hides navigation and irrelevant UI elements

3.7 Audit Log Module

Description

Records all significant user actions across the system for security and accountability.

Features

- Logs: user logins, logouts, record uploads, patient approvals, rejections, edits, and deletions
- Each entry records: action type, user ID, timestamp, and description
- Administrator can search and filter audit log by user or action type
- Audit entries cannot be modified or deleted by any user role

3.8 Contact / Support Module

Description

Provides patients and staff with a form to submit support enquiries to the Administrator.

Inputs

- Name, email, subject, message body

Outputs

- Message stored in support_messages table
- Confirmation message displayed to the user

4. Non-Functional Requirements

4.1 Performance Requirements

- Page load time should not exceed 3 seconds on a standard LAN connection
- Patient search results should be returned within 2 seconds
- File uploads should support files up to 20 MB per attachment
- The system should support at least 20 concurrent users on a LAN without degradation

4.2 Security Requirements

- All passwords stored using `password_hash()`
- PHP execution is blocked inside the uploads/ directory via `.htaccess`
- File uploads are validated by both file extension
- Session-based authentication enforced on every protected page
- Patient accounts require Administrator approval before login is permitted

4.3 Usability Requirements

- The interface must be fully mobile-responsive across all screen sizes
- Navigation must be consistent across all user roles
- Patient age must be automatically calculated and displayed — no manual entry required
- The system must adhere to the GH+ red, white, and black design theme

4.5 Maintainability Requirements

- The good PHP architecture
- Database schema is documented
- Code is organized for future modification
- XAMPP environment allows easy server management via the Control Panel

4.6 Portability Requirements

- The system runs on any OS that supports XAMPP
- The frontend is compatible with all modern browsers (Chrome, Firefox, Edge, Safari)
- No client-side installation is required — access via browser only

zx5. System Requirements

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	50 GB free disk space for database and file uploads
Network	Local Area Network (LAN) for multi-user access
Display	1366 x 768 resolution or higher
Input Devices	Keyboard, mouse, and optional document scanner

Software Requirements

6. Database Design

6.1 Database Overview

The GH+ system uses a single MySQL database named gh_plus. The schema is fully normalized .

6.2 Core Tables

Table Name	Description
users	Stores all user accounts (patients, doctors, admin) with role, NIC, password hash, and approval status
patient_details	Stores patient-specific data: blood type, known diseases, additional notes — linked 1:1 to users
staff_details	Stores medical staff data: position, hospital, department, SLMC registration number — linked 1:1 to users
medical_records	Stores each medical visit record: diagnosis, prescription, doctor notes — links patient and doctor via foreign keys
medical_documents	Stores metadata for file attachments (images, PDFs, audio, video) linked to medical records; actual files stored in uploads/
support_messages	Stores patient/staff support enquiries from the Contact Us page
audit_log	Records all significant user actions: logins, uploads, approvals, deletions — references users table

6.3 Relationships

- users is the central entity; patient_details and staff_details extend it via one-to-one relationships
- medical_records links patients and doctors through foreign keys to the users table
- medical_documents is linked to medical_records via foreign key
- audit_log references users to track who performed each action
- All foreign keys enforce referential integrity with CASCADE DELETE where appropriate

7. User Interface Requirements

7.1 Design Theme

GH+ follows a red, white, and black colour scheme reflecting the GH+ brand identity. CSS custom properties maintain consistent theming. Layouts use CSS Grid and Flexbox for full responsiveness.

7.2 Screens Included

Screen	Accessible By
Login Page	All users
Patient Registration Page	Public (unauthenticated)
Patient Dashboard	Patient
Patient Medical History View	Patient
Medical Record PDF Export	Patient
Medical Staff Dashboard	Doctor / Medical Staff
Patient Search Results	Doctor / Medical Staff
New Medical Record Upload Form	Doctor / Medical Staff
Administrator Dashboard	Administrator
User Approval Panel	Administrator
User Management Panel	Administrator
System Reports	Administrator

8. Advantages of the System

- Eliminates paper-based records — prevents record loss and damage across hospital transfers
- Centralized patient data enables faster diagnosis in emergency situations
- Role-based access ensures data privacy and compliance with medical confidentiality standards
- Reduces duplicate testing by making complete medical history available system-wide
- Empowers patients with transparent, self-service access to their own health data

- Audit logging ensures full accountability for all actions within the system
- Cost-effective: built entirely on free, open-source technologies (PHP, MySQL, XAMPP)
- Mobile-responsive — accessible on smartphones without any app installation
- PDF download feature allows patients to share records with private practitioners

9. Future Enhancements

- Integration with a national digital health identity system
- SMS or email notifications for appointment reminders and record updates
- Telemedicine module allowing remote doctor consultations
- AI-assisted diagnosis suggestion based on symptom keywords in records
- Multi-language support (Sinhala and Tamil) for broader accessibility
- Cloud migration option for inter-hospital data sharing across different LAN environments
- Barcode/QR-based patient identification at reception
- Laboratory result integration and automated reporting

10. Conclusion

Government Hospital Plus (GH+) is a purpose-built, secure, and cost-effective digital medical records management system designed specifically for Sri Lanka's government hospital network. By replacing paper-based processes with a centralized, role-based web application, GH+ addresses critical inefficiencies including record loss, duplicate entries, medication errors, and limited patient access to personal health data.

The system's use of proven, locally deployable technologies — PHP, MySQL, and XAMPP — ensures that GH+ can be implemented without dependency on expensive cloud infrastructure, making it a practical and sustainable solution for government hospital environments across Sri Lanka. The comprehensive security framework, including BCrypt password hashing, prepared statements, session management, MIME type validation, and full audit logging, ensures that patient data remains confidential, traceable, and protected at all times.

GH+ represents a meaningful step toward modernizing healthcare record management in Sri Lanka, improving patient outcomes, reducing administrative burden on medical staff, and building a foundation for future digital health integration.